

BUSINESS INTELLIGENCE PROJECT

Livspace Interior Design

Project & Operations Analytics Dashboard

MySQL • Power Query • DAX • Microsoft Power BI

Presented by

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Project Overview

An enterprise-level BI case study for an interior design & renovation business

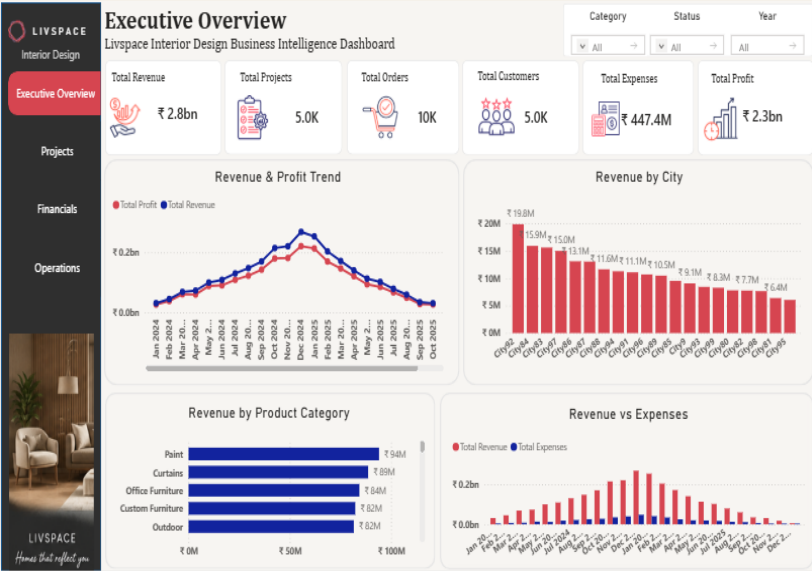
This project analyzes and monitors the performance of an interior design and home renovation business by integrating data across customers, designers, employees, projects, orders, payments, suppliers, inventory, expenses, products, and reviews — transforming fragmented operational data into insights that support management decisions.

Overall Objective



Key Objectives

- Track project progress and flag delayed projects
- Analyze customer behavior, ratings & feedback
- Evaluate designer and employee productivity
- Monitor inventory, suppliers & procurement



Executive Overview — live snapshot

15 Source Tables

80,240 Total Records

4 Dashboard Pages

Industry & Problem Statement

Interior Design & Home Renovation

Industry

Interior design & home renovation services — modular kitchens, wardrobes, furniture, renovation, material sourcing, and turnkey project execution across multiple cities.

Problem Statement

Operating across multiple cities generates large volumes of data across customers, projects, employees, orders, payments, inventory and suppliers — leaving management without a centralized view of business performance.

Resolved By

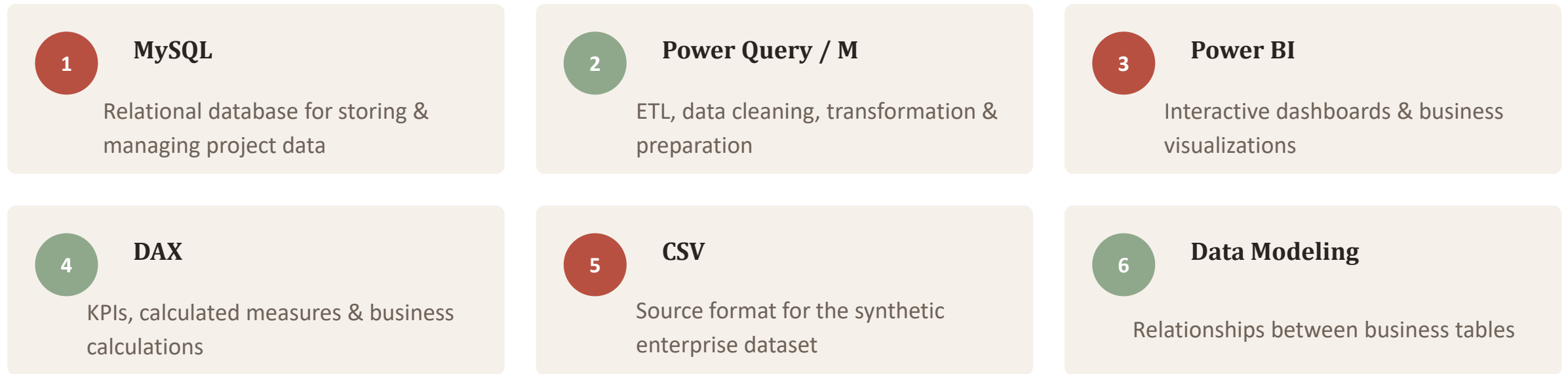
A centralized BI dashboard that integrates, cleans, analyzes, and visualizes data end-to-end.

Major Problems

- ! Difficulty monitoring overall business performance
- ! Tracking project progress & delayed projects
- ! Limited visibility into designer/employee productivity
- ! Monitoring revenue, expenses & profitability
- ! Understanding customer satisfaction & feedback
- ! Inefficient inventory & supplier monitoring
- ! Identifying high-performing cities & categories
- ! Data-quality issues: missing values, inconsistencies

Tools & Technologies

The technology stack powering this BI solution



Technology Workflow



Dataset Introduction & Source

A synthetic, enterprise-scale relational dataset

The dataset was created specifically for this educational and portfolio project. It simulates realistic business operations of an interior design company in a normalized Third Normal Form (3NF) relational structure — and does not contain confidential or proprietary Livspace data.

15

Tables

80,240

Total Records

3NF

Normalization

Covers customers, employees, designers, projects, products, orders, payments, finance, inventory, suppliers, and customer satisfaction data.

Table	Records
Cities	250
Customers	5,000
Employees	300
Designers	150
Suppliers	500
Product_Categories	40
Products	2,000
Projects	5,000
Project_Tasks	20,000
Orders	10,000
Order_Items	25,000
Payments	7,000
Expenses	3,000
Inventory	1,500
Customer_Reviews	500

MySQL Database & Power BI Connection

Building the relational foundation and linking it to Power BI

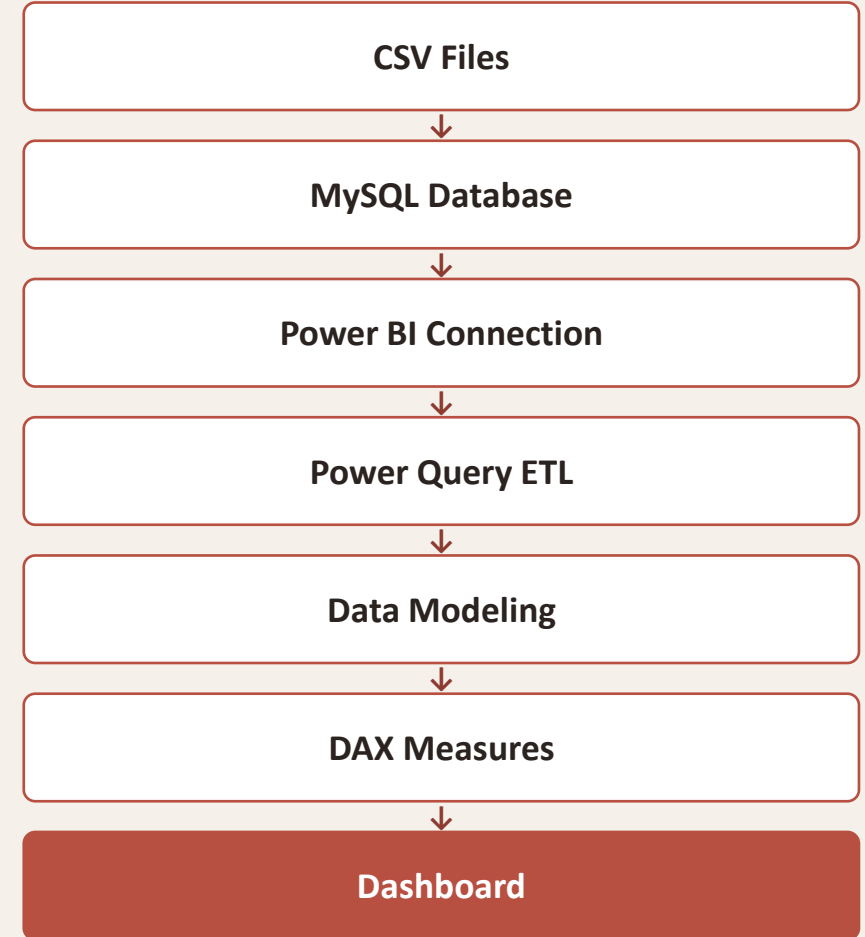
MySQL — Database & Tables

CSV files are imported into MySQL as separate relational tables. Primary-key and foreign-key relationships are established to build a structured database supporting analysis of customers, projects, orders, sales, payments, expenses, products, inventory, suppliers, and reviews.

Connecting MySQL to Power BI

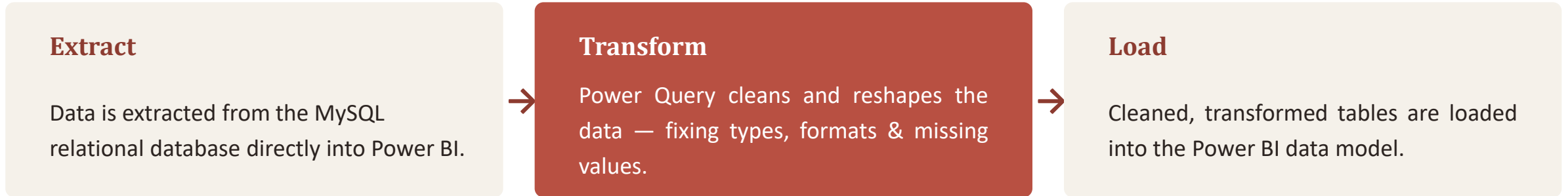
- 1 Open Power BI Desktop → Home → Get Data
- 2 Select “MySQL database” as the source
- 3 Enter MySQL server & database details
- 4 Select the required tables
- 5 Choose “Transform Data” to enter Power Query

Database Flow



ETL — Loading Data & Data Cleaning

Extract → Transform → Load, powered by Power Query



Data Cleaning Tasks (Power Query)

- ✓ Handling missing values
- ✓ Removing error rows
- ✓ Correcting inconsistent date formats
- ✓ Correcting date & numeric data types
- ✓ Creating required calculated columns
- ✓ Creating project task duration values
- ✓ Standardizing categorical values & capitalization
- ✓ Handling missing ratings & contact info
- ✓ Calculating missing discount values in Order_Items

Basic DAX Measures

Core KPIs powering every dashboard page

Total Revenue

Sums total revenue generated

Completed / Delayed Projects

Counts projects by status

Recommendation Rate %

Customers who would recommend the service

Profit Margin %

Profit as a percentage of revenue

Total Inventory Value

Value of available inventory

Products Requiring Reorder

Combines low-stock & out-of-stock items

Total Profit

Revenue minus total expenses

Average Customer Rating

Calculates average customer rating

Average Project Duration

Average task / project duration

Payment Collection %

Measures payment collection efficiency

Low Stock / Out-of-Stock

Products at or below reorder level

Active Suppliers

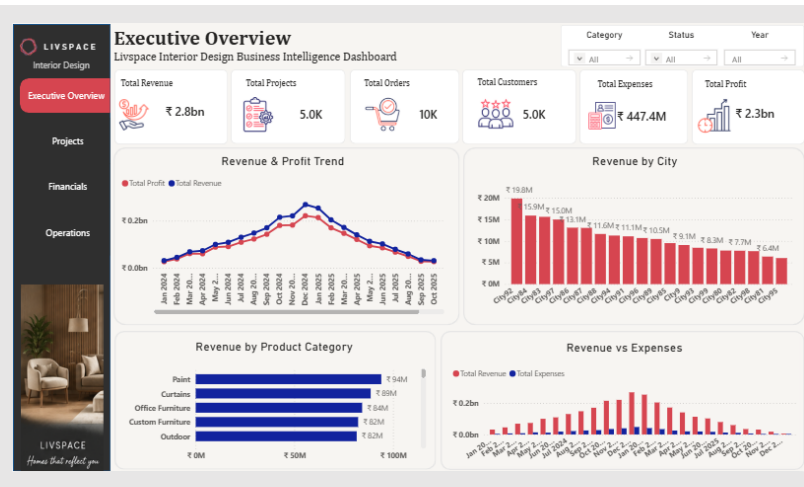
Suppliers linked to available inventory

Dashboard Design — Pages Overview

Four consolidated Power BI pages, one for each business lens

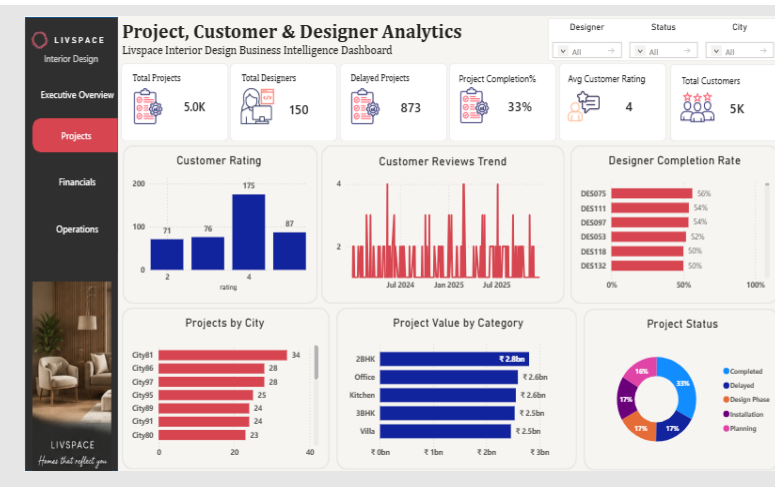
Executive Overview

Overall business performance



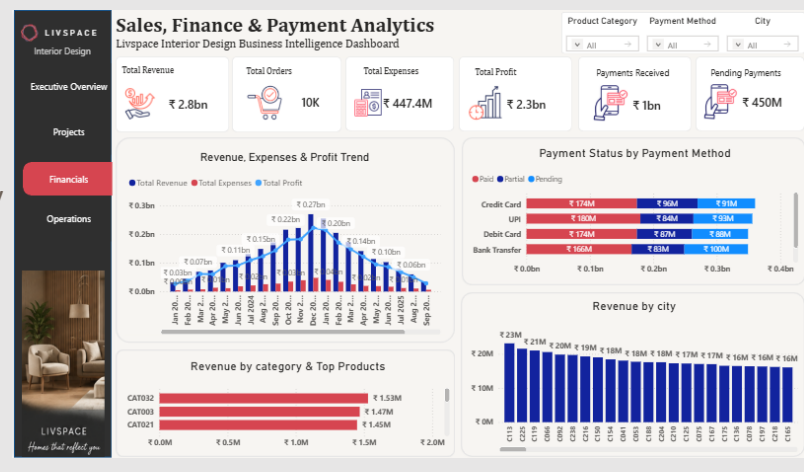
Project, Customer & Designer

Execution & satisfaction



Sales, Finance & Payment

Revenue & profitability



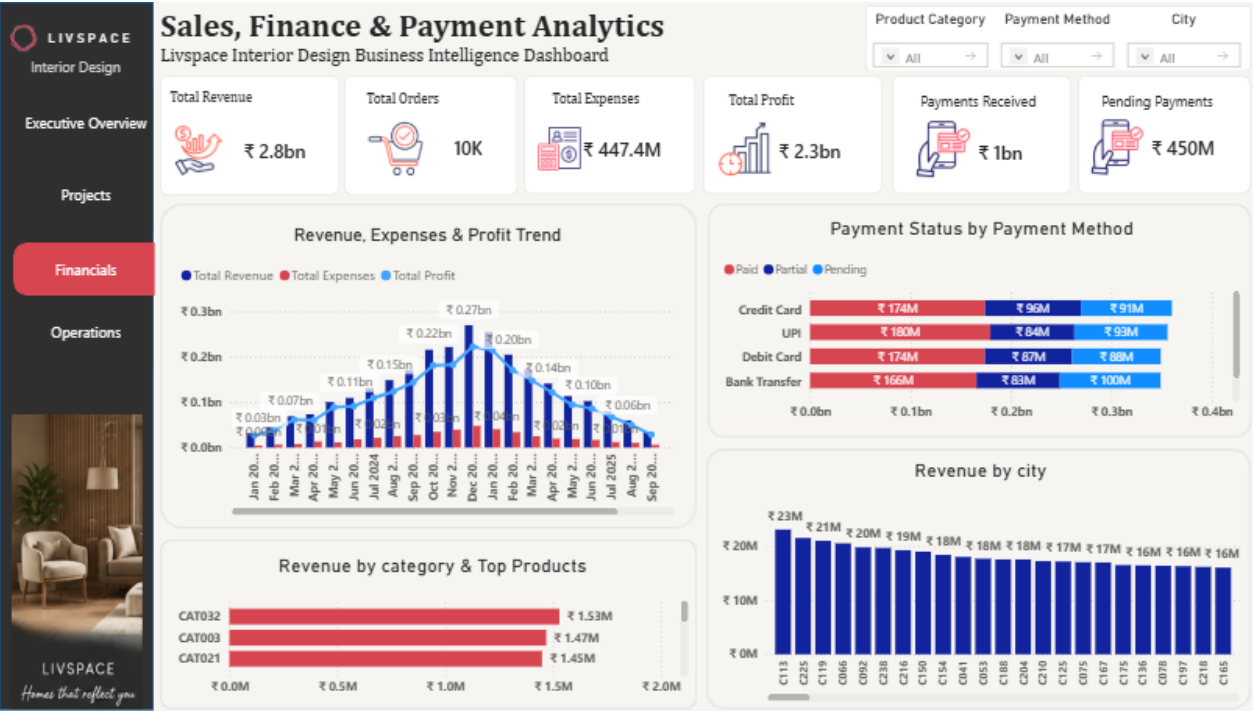
Operations, Inventory & Supplier

Stock & procurement



Dashboard Pages, Visuals & Filters

Interactive slicers keep every page connected and drillable

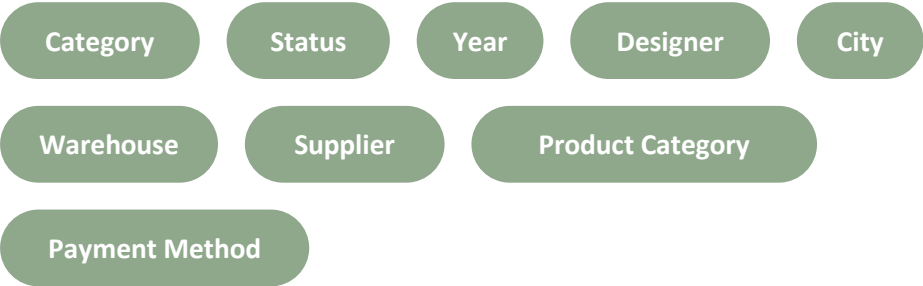


Sales, Finance & Payment Analytics — example page

Visuals Used Across Pages

KPI cards · trend lines · bar & column charts · donut charts · tree maps · tables & matrix visuals

Filters & Slicers



Business Questions Answered

- How is the business performing overall?
- Are projects delivered efficiently with high satisfaction?
- How well is revenue collected & profitability maintained?
- Is inventory & supplier capacity right-sized for projects?

Key Insights

What the dashboard enables management to do



Monitor overall revenue, profit, expenses, orders & project performance



Compare revenue across cities and product categories



Identify delayed projects & evaluate completion performance



Compare designer performance and completion rates



Understand customer ratings and review trends



Track revenue, expense & profit trends over time



Analyze payment collection and pending payments



Flag low-stock and out-of-stock products needing attention



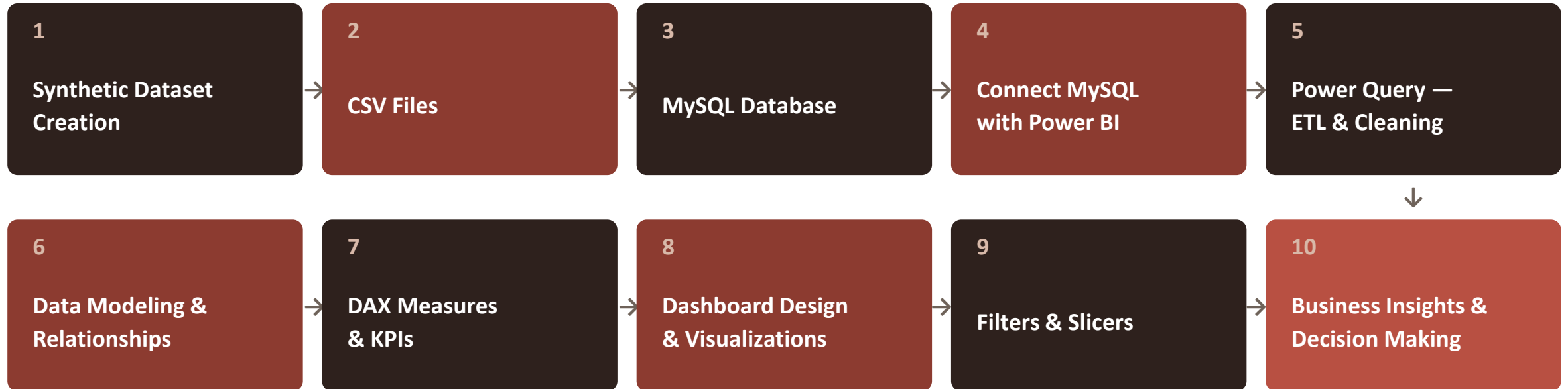
Analyze inventory by warehouse, category, supplier & product



Spot suppliers and products driving business operations

Project Workflow

From synthetic data creation to business decisions



Workflow Summary

Data → ETL → Data Model → DAX → Visualization → Insights → Decision Making

GitHub Repository

<https://github.com/Gayathri0109/livspace-interior-analytics-dashboard.git>